

Large Language Models

Prompting for Zero-Shot and Few-Shot Learning

Mohammad Hossein Rohban

Fall 2023

Courtesy: Most of the slides are adopted from the course COS 597G and the paper “Making Pre-trained Language Models Better Few-shot Learners” by Gao et al.

What is Zero-Shot Learning?

Foundational Model
General learner

- Zero-Shot Learning (ZSL) [2009-]

- **Unseen** test sample **classes** (or tasks) during training

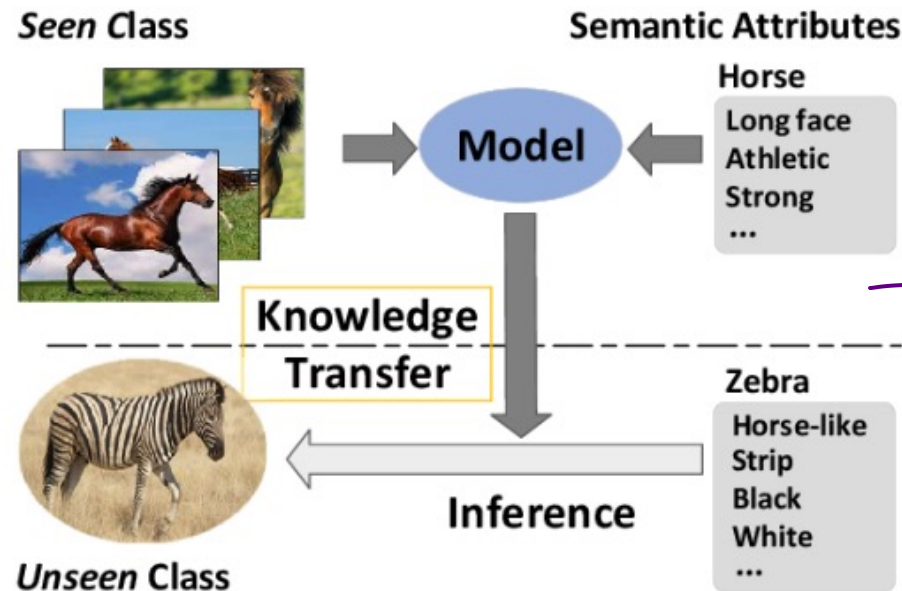
- Has to **associate** observed and non-observed classes

- **Auxiliary information** is used to make this happen

- e.g. a model trained to **recognize horses** along with **textual info** of how each animal looks like 🐾 can classify **zebras** too!

مفهوم حیران

مدل آرد مفهوم را خوب یاد گرفته باشد



مفهوم

اطلاعات
کمی درست

اما تا حالا ازش دیتا ندیده
اسم کرد خرد هست

ZSL (cont.)

- T0: An encoder-decoder model
 - 16x smaller than GPT-3
 - Can generalize to unseen NLP tasks
 - Explicit multi-task learning to achieve ZSL.
 - Map any NLP task into a readable prompt.
 - Fine-tuned the T5 model on multi-task training dataset.
 - <https://bigscience.huggingface.co/blog/t0>

ملاس جدید
نک جدید

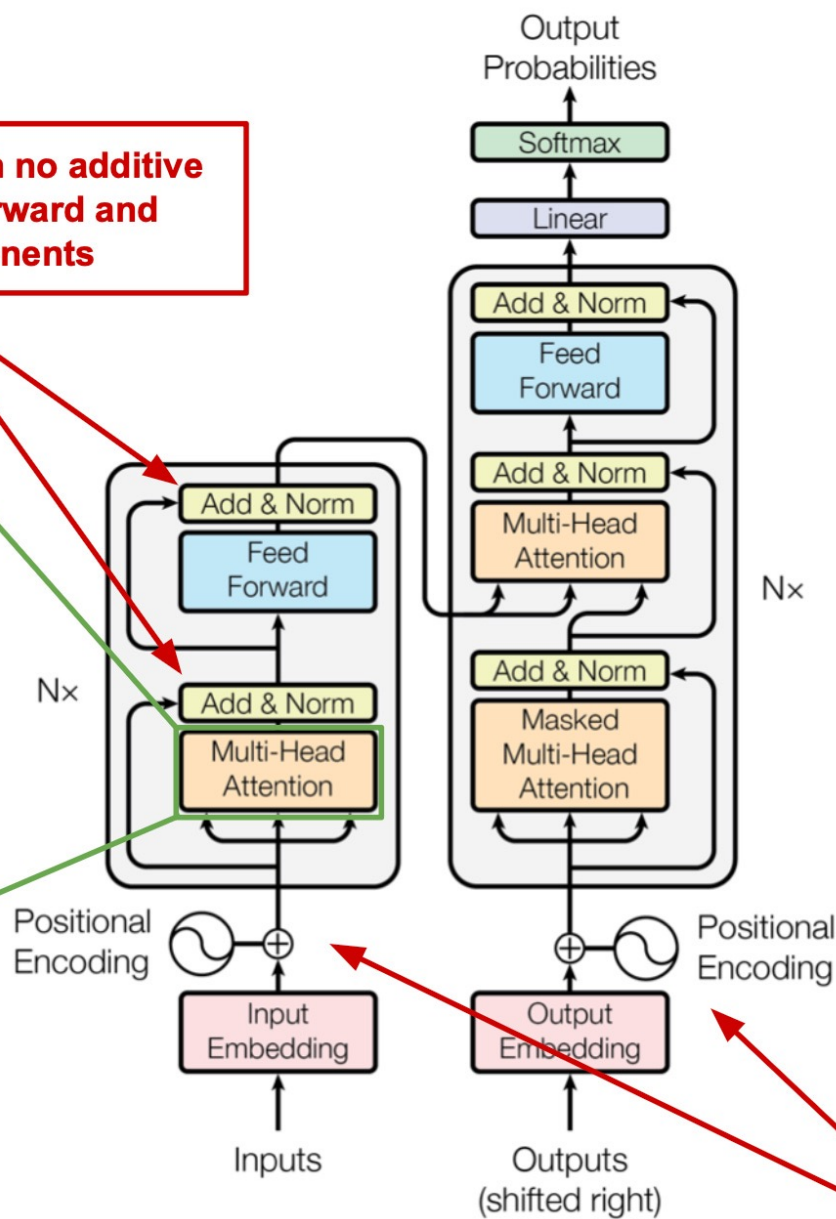
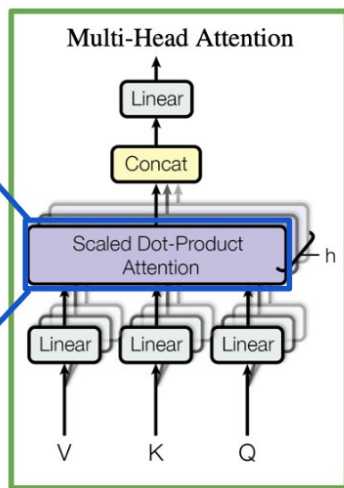
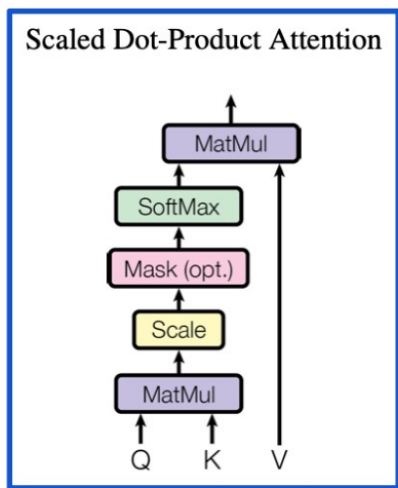
← میدان باعث مرشد شبکه خنثی به

T5 Model

بایاس تکیه کنه تا وزن 4

خونک این ترم

Apply LayerNorm (with no additive bias) before feed forward and attention components



Use a simplified positional encoding scheme

تبدیل task به QA

Summarization

The picture appeared on the wall of a Poundland store on Whymark Avenue [...] How would you rephrase that in a few words?

Sentiment Analysis

Review: We came here on a Saturday night and luckily it wasn't as packed as I thought it would be [...] On a scale of 1 to 5, I would give this a

Question Answering

I know that the answer to "What team did the Panthers defeat?" is in "The Panthers finished the regular season [...]". Can you tell me what it is?

T0

Graffiti artist Banksy is believed to be behind [...]

4

Arizona Cardinals

Multi-task training

Zero-shot generalization

Natural Language Inference

Suppose "The banker contacted the professors and the athlete". Can we infer that "The banker contacted the professors"?

دیتا ریکوردر
مدل کدچیت

Decoder-Encoder

دلی واره shot حزب عمل

نست به
GPT

Yes

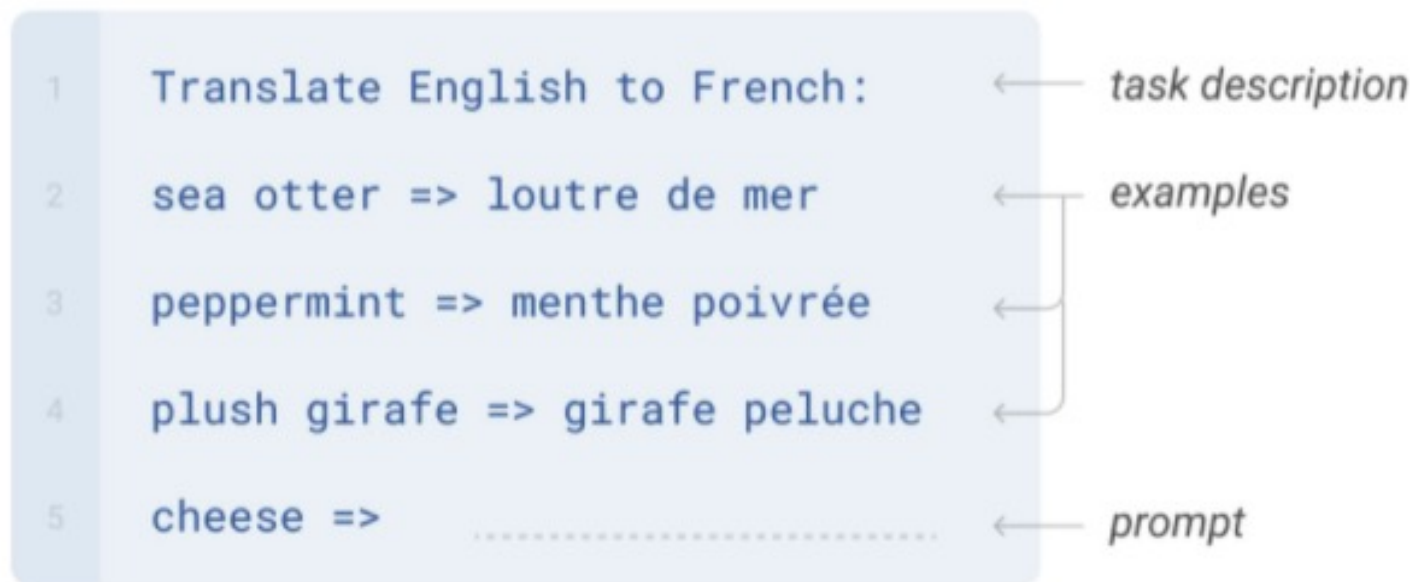
اكتبت
↑
شمار

- وزن ل بدون عین

Few-Shot Learning

- in context learning -

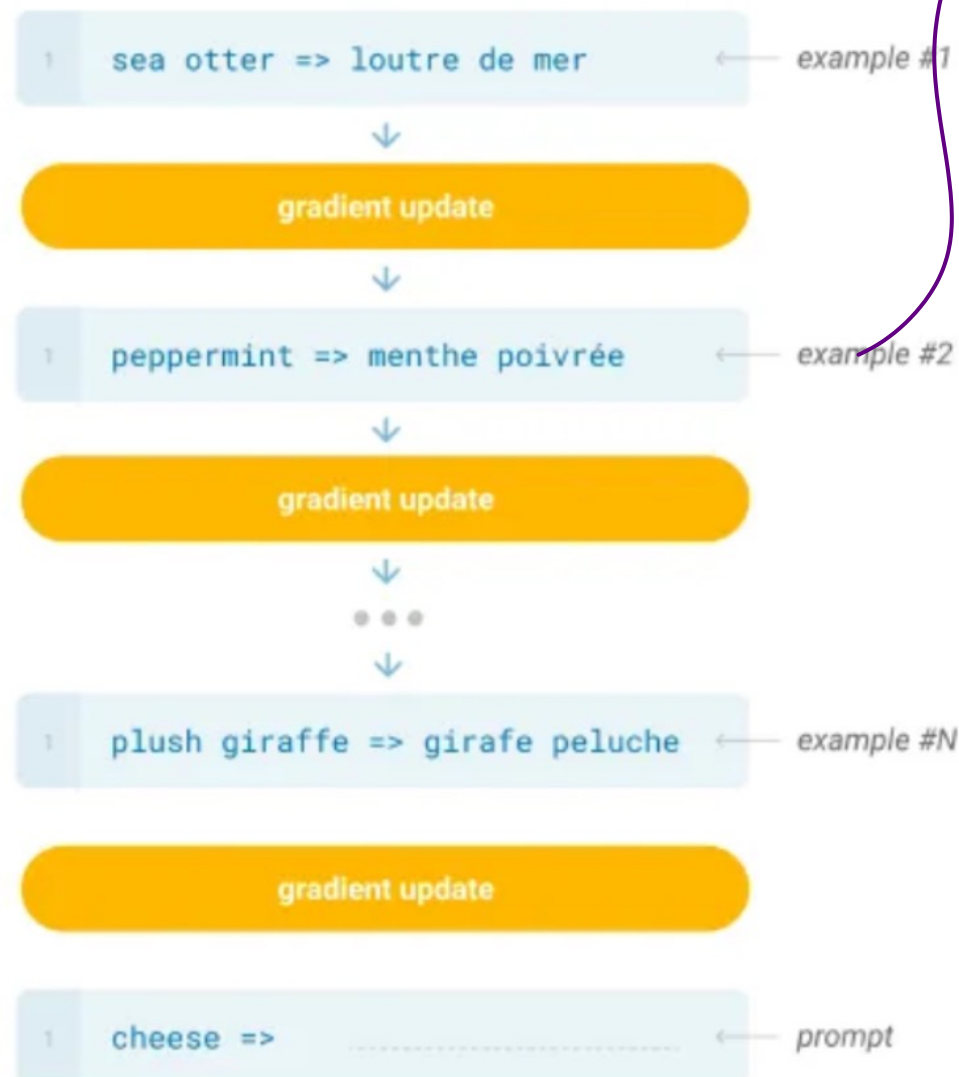
- Including few examples of test task at inference time.



Traditional fine-tuning (not used for GPT-3)

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



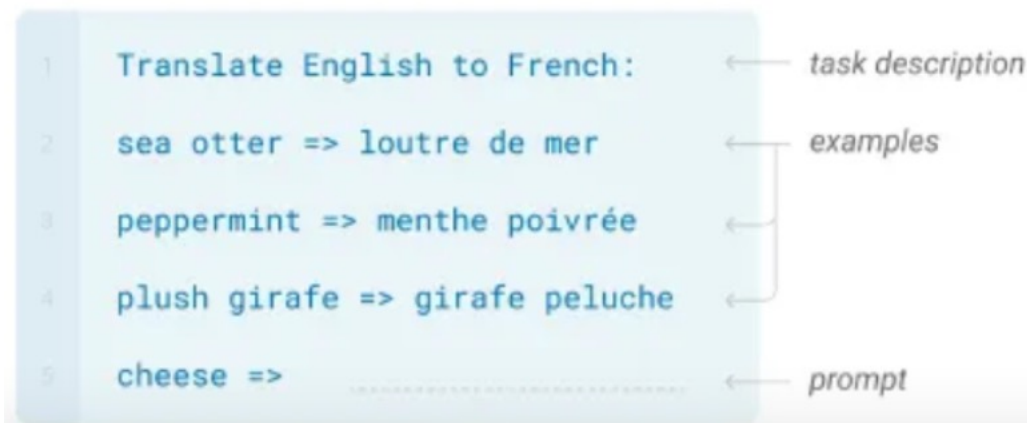
Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.



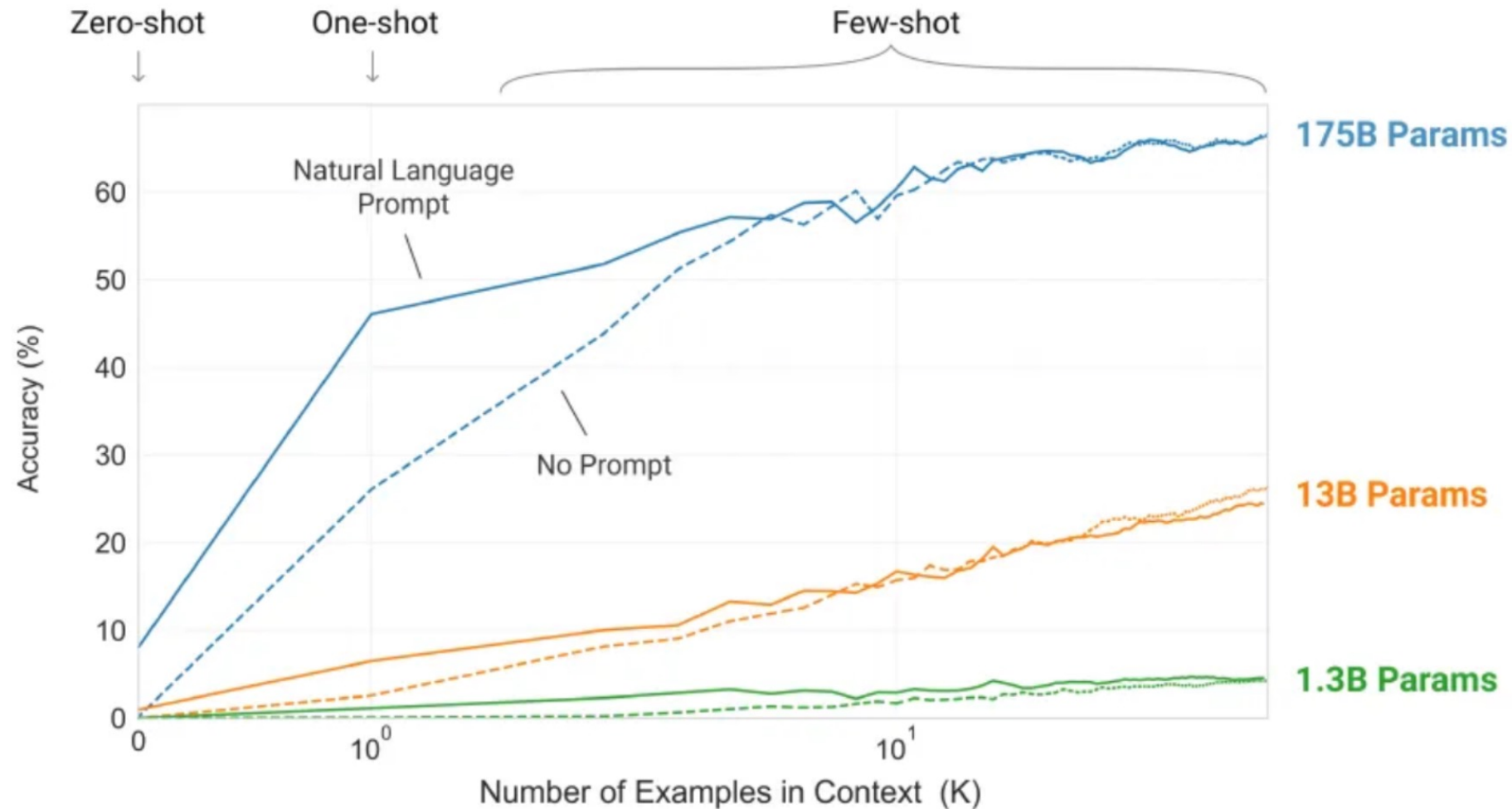
Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



باید سایر مدل به اندازه کافی بزرگ باشد تا به چنین چیزی *emerge* بکند

LLMs have ZSL and FSL capabilities



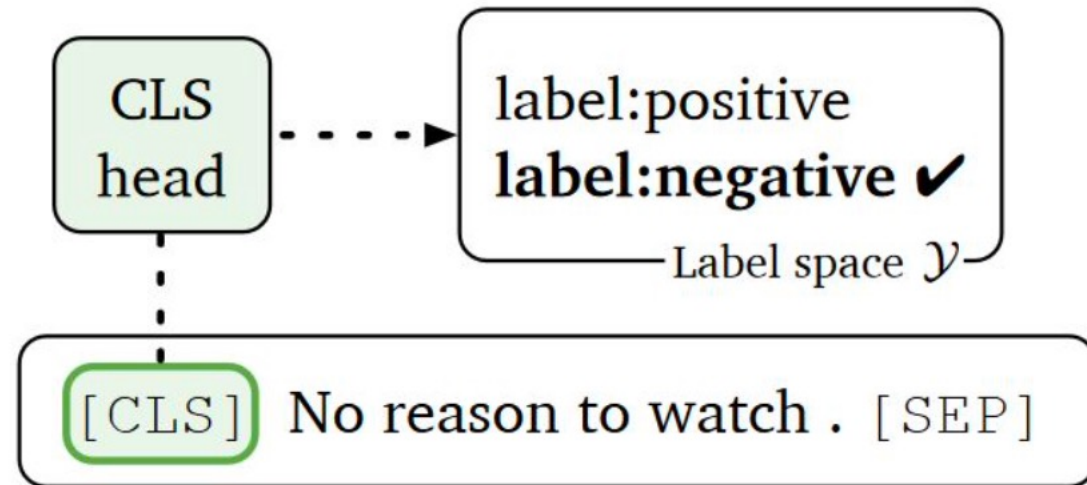
improved ability to learn a task from contextual information

few-shot learning

Let's have a discussion

- Don't have the luxury of deploying a 100-B parameter.
 - All we can afford is a pre-trained 100-M parameter model.
- Have only a **couple of labeled examples** from the target task.
 - Let's say sentiment analysis of movies.
- How to go about this?

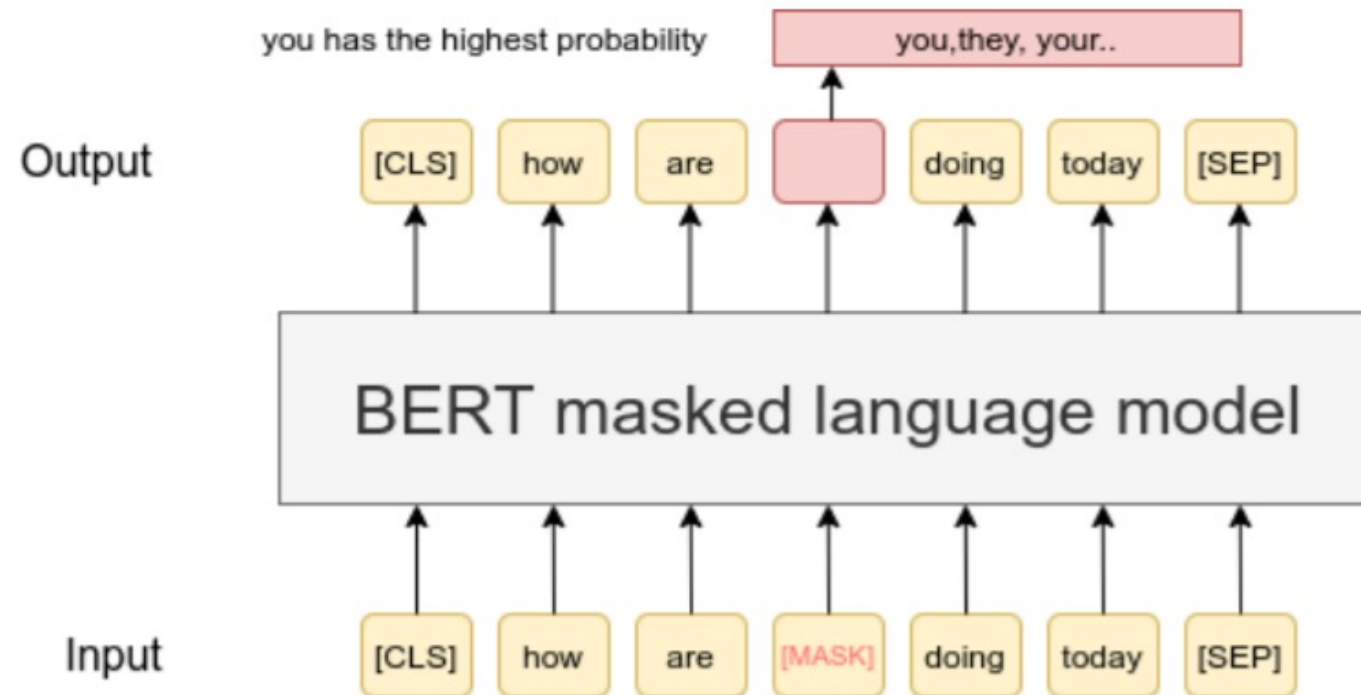
1st Solution: Head-based Fine-Tuning of a MLM



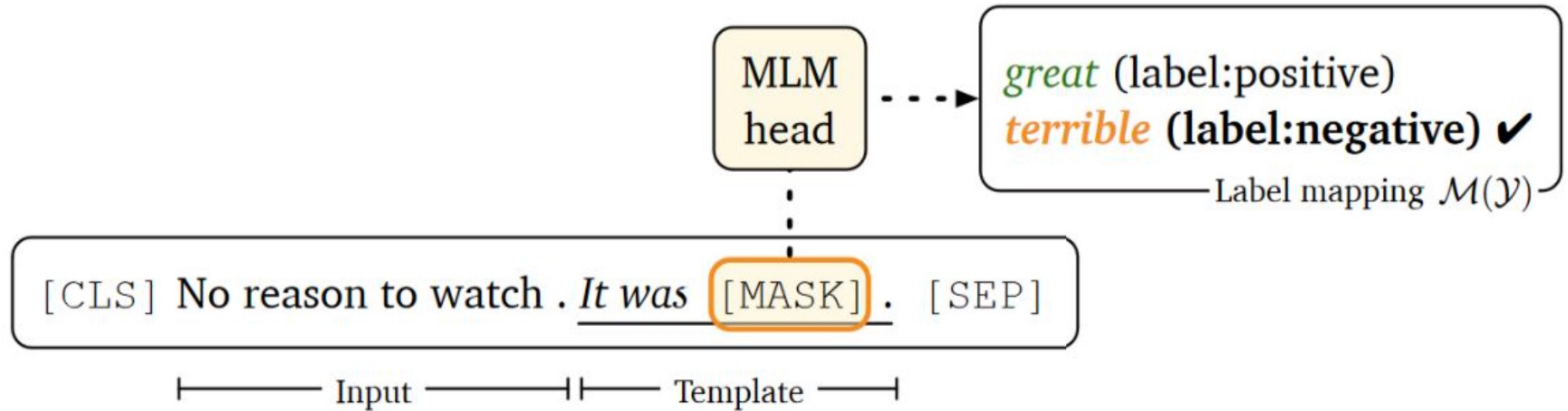
- How many trainable parameters are involved?
- $\text{hidden_size} \times \text{num_classes}$
- Does it work well when given only ~ 10 training samples?

What else we can do? Let's discuss.

- ... which better suits the FSL setup?
- Utilizing the **masked token prediction** capability of the BERT.

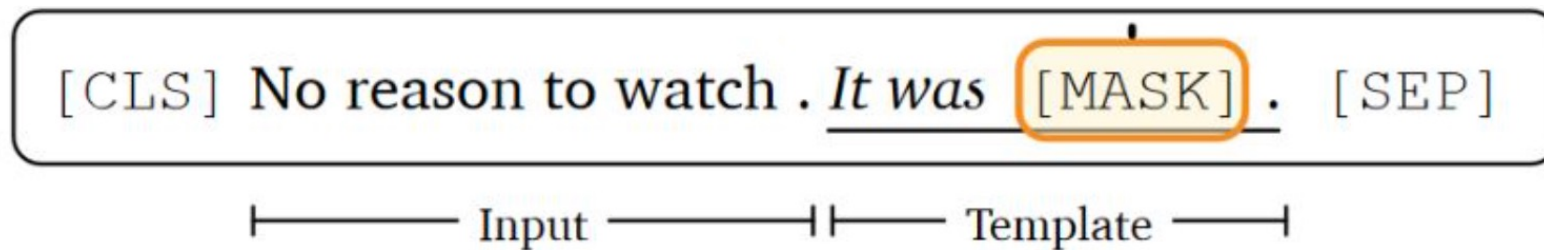


Prompt-based Fine-Tuning

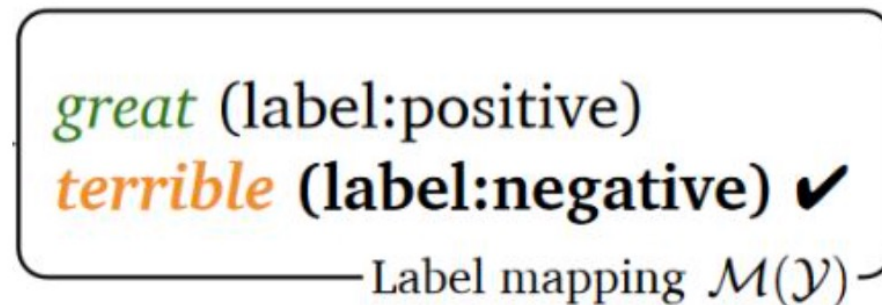


Prompt-based Fine-Tuning (cont.)

- Step 1: **Formulate** the task into a masked token prediction through a **prompt template**:



- Step 2: Choose a **label-word mapping** M .



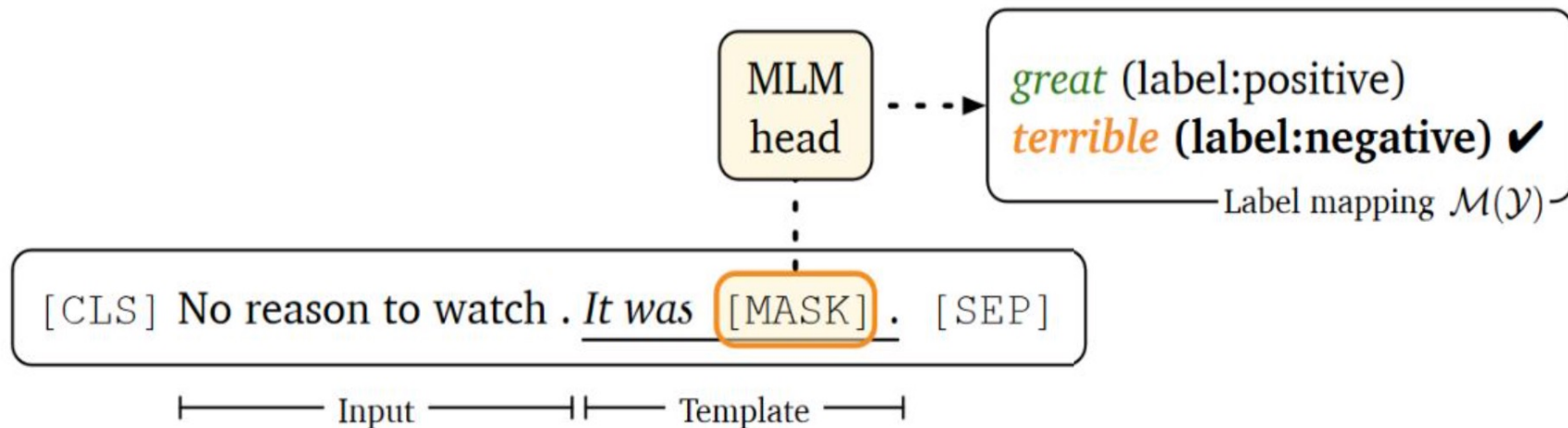
Prompt-based Fine-Tuning (cont.)

- Step 3: **Fine-tune** the LM to fill in the correct word

$$p(y \mid x_{\text{in}}) = p([\text{MASK}] = \mathcal{M}(y) \mid x_{\text{prompt}}) \\ = \frac{\exp(\mathbf{w}_{\mathcal{M}(y)} \cdot \mathbf{h}_{[\text{MASK}]})}{\sum_{y' \in \mathcal{Y}} \exp(\mathbf{w}_{\mathcal{M}(y')} \cdot \mathbf{h}_{[\text{MASK}]})},$$

فقدیم این احتمال کلیدی
که map داریم جمع می‌زنند

! نه به لایه!



Regression Problem

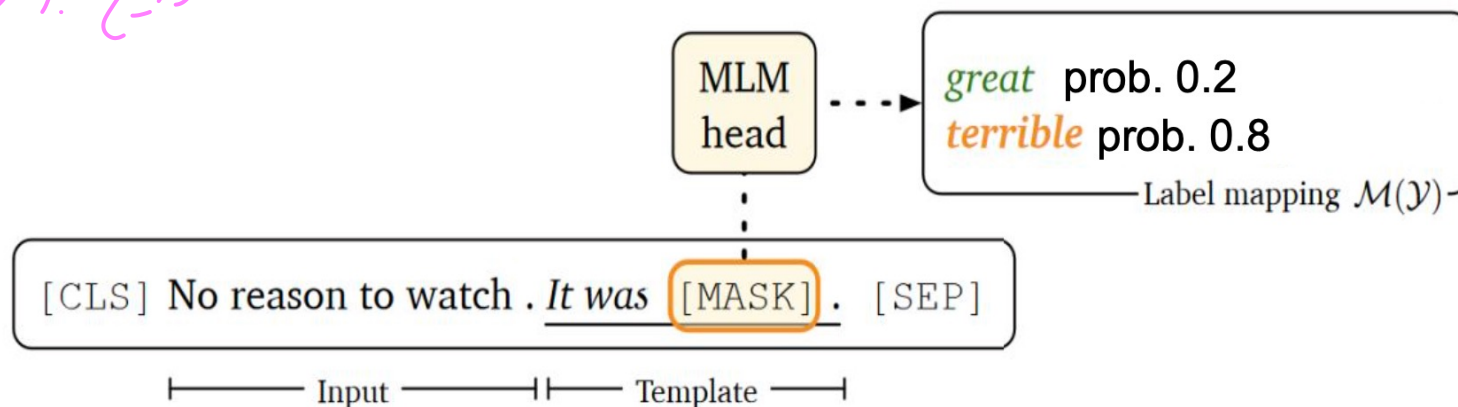
تبدیل احتمال به یک عدد پیوسته

- Regression: interpolating between two extremes

$$y = v_l \cdot p(y_l | x_{in}) + v_u \cdot p(y_u | x_{in}) \Rightarrow \text{تبدیل خطی}$$

- The LM is fine-tuned to minimize the KL-divergence between the **inferred** $P(y_u | x_{in})$ and $(y - v_l)/(v_u - v_l)$ the **observed target**.

در ستا توزیع بر نوزی



Predicted $y = 0.2$

Evaluation Datasets

Category	Dataset	$ \mathcal{Y} $	L	#Train	#Test	Type	Labels (classification tasks)
single-sentence	SST-2	2	19	6,920	872	sentiment	positive, negative
	SST-5	5	18	8,544	2,210	sentiment	v. pos., positive, neutral, negative, v. neg.
	MR	2	20	8,662	2,000	sentiment	positive, negative
	CR	2	19	1,775	2,000	sentiment	positive, negative
	MPQA	2	3	8,606	2,000	opinion polarity	positive, negative
	Subj	2	23	8,000	2,000	subjectivity	subjective, objective
	TREC	6	10	5,452	500	question cls.	abbr., entity, description, human, loc., num.
	CoLA	2	8	8,551	1,042	acceptability	grammatical, not_grammatical
استلزام منطقي sentence-pair	MNLI	3	22/11	392,702	9,815	NLI	entailment, neutral, contradiction
	SNLI	3	14/8	549,367	9,842	NLI	entailment, neutral, contradiction
	QNLI	2	11/30	104,743	5,463	NLI	entailment, not_entailment
	RTE	2	49/10	2,490	277	NLI	entailment, not_entailment
	MRPC	2	22/21	3,668	408	paraphrase	equivalent, not_equivalent
	QQP	2	12/12	363,846	40,431	paraphrase	equivalent, not_equivalent
	STS-B	$\mathcal{R}$	11/11	5,749	1,500	sent. similarity	-

Examples

- SST-2: sentiment analysis.
- e.g. S_1 = “The movie is ridiculous”. Label: negative.
- Manual prompt:

Template	Label words
$\langle S_1 \rangle$ It was [MASK] .	great/terrible

Examples (cont.)

- SNLI: Natural Language Inference
- S_1 = “A soccer game with multiple males playing”. S_2 = “Some men are playing sport”. Label: Entailment.
- Manual prompt:

Template	Label words
$\langle S_1 \rangle$? [MASK] , $\langle S_2 \rangle$	Yes/Maybe/No

Few-shot Learning & Evaluation Protocol

- Training dataset: $K=16$ examples per class.
- Dev dataset: same size as training dataset.
- Performance measured across 5 random splits of {train, dev} set.

اصل از ادن $\text{prior} - \text{know}$ نمودگی استفاده کنه! چون مقدار متنش بیشتره

Results

	SST-2 (acc)	SST-5 (acc)	MR (acc)	CR (acc)	MPQA (acc)	Subj (acc)	TREC (acc)	CoLA (Matt.)
Majority [†]	50.9	23.1	50.0	50.0	50.0	50.0	18.8	0.0
Prompt-based zero-shot [‡]	83.6	35.0	80.8	79.5	67.6	51.4	32.0	2.0
“GPT-3” in-context learning	84.8 (1.3)	30.6 (0.9)	80.5 (1.7)	87.4 (0.8)	63.8 (2.1)	53.6 (1.0)	26.2 (2.4)	-1.5 (2.4)
Fine-tuning	81.4 (3.8)	43.9 (2.0)	76.9 (5.9)	75.8 (3.2)	72.0 (3.8)	90.8 (1.8)	88.8 (2.1)	33.9 (14.3)
Prompt-based FT (man)	92.7 (0.9)	47.4 (2.5)	87.0 (1.2)	90.3 (1.0)	84.7 (2.2)	91.2 (1.1)	84.8 (5.1)	9.3 (7.3)
+ demonstrations	92.6 (0.5)	50.6 (1.4)	86.6 (2.2)	90.2 (1.2)	87.0 (1.1)	92.3 (0.8)	87.5 (3.2)	18.7 (8.8)
Prompt-based FT (auto)	92.3 (1.0)	49.2 (1.6)	85.5 (2.8)	89.0 (1.4)	85.8 (1.9)	91.2 (1.1)	88.2 (2.0)	14.0 (14.1)
+ demonstrations	93.0 (0.6)	49.5 (1.7)	87.7 (1.4)	91.0 (0.9)	86.5 (2.6)	91.4 (1.8)	89.4 (1.7)	21.8 (15.9)
Fine-tuning (full) [†]	95.0	58.7	90.8	89.4	87.8	97.0	97.4	62.6
	MNLI (acc)	MNLI-mm (acc)	SNLI (acc)	QNLI (acc)	RTE (acc)	MRPC (F1)	QQP (F1)	STS-B (Pear.)
Majority [†]	32.7	33.0	33.8	49.5	52.7	81.2	0.0	-
Prompt-based zero-shot [‡]	50.8	51.7	49.5	50.8	51.3	61.9	49.7	-3.2
“GPT-3” in-context learning	52.0 (0.7)	53.4 (0.6)	47.1 (0.6)	53.8 (0.4)	60.4 (1.4)	45.7 (6.0)	36.1 (5.2)	14.3 (2.8)
Fine-tuning	45.8 (6.4)	47.8 (6.8)	48.4 (4.8)	60.2 (6.5)	54.4 (3.9)	76.6 (2.5)	60.7 (4.3)	53.5 (8.5)
Prompt-based FT (man)	68.3 (2.3)	70.5 (1.9)	77.2 (3.7)	64.5 (4.2)	69.1 (3.6)	74.5 (5.3)	65.5 (5.3)	71.0 (7.0)
+ demonstrations	70.7 (1.3)	72.0 (1.2)	79.7 (1.5)	69.2 (1.9)	68.7 (2.3)	77.8 (2.0)	69.8 (1.8)	73.5 (5.1)
Prompt-based FT (auto)	68.3 (2.5)	70.1 (2.6)	77.1 (2.1)	68.3 (7.4)	73.9 (2.2)	76.2 (2.3)	67.0 (3.0)	75.0 (3.3)
+ demonstrations	70.0 (3.6)	72.0 (3.1)	77.5 (3.5)	68.5 (5.4)	71.1 (5.3)	78.1 (3.4)	67.7 (5.8)	76.4 (6.2)
Fine-tuning (full) [†]	89.8	89.5	92.6	93.3	80.9	91.4	81.7	91.9

بجای

بجای

جایی

جایی

Table 3: Our main results using RoBERTa-large. [†]: full training set is used (see dataset sizes in Table B.1); [‡]: no training examples are used; otherwise we use $K = 16$ (per class) for few-shot experiments. We report mean (and standard deviation) performance over 5 different splits (§3). Majority: majority class; FT: fine-tuning; man: manual prompt (Table 1); auto: automatically searched templates (§5.2); “GPT-3” in-context learning: using the in-context learning proposed in Brown et al. (2020) with RoBERTa-large (no parameter updates).

Effect of Word-Class Mapping

Template	Label words	Accuracy
SST-2 (positive/negative)		mean (std)
$\langle S_1 \rangle$ It was [MASK] .	great/terrible	92.7 (0.9)
$\langle S_1 \rangle$ It was [MASK] .	good/bad	92.5 (1.0)
$\langle S_1 \rangle$ It was [MASK] .	cat/dog	91.5 (1.4)
$\langle S_1 \rangle$ It was [MASK] .	dog/cat	86.2 (5.4)
$\langle S_1 \rangle$ It was [MASK] .	terrible/great	83.2 (6.9)
Fine-tuning	-	81.4 (3.8)

Effect of the Prompt Template

SNLI (entailment/neutral/contradiction)		mean (std)
$\langle S_1 \rangle$? [MASK] , $\langle S_2 \rangle$	Yes/Maybe/No	77.2 (3.7)
$\langle S_1 \rangle$. [MASK] , $\langle S_2 \rangle$	Yes/Maybe/No	76.2 (3.3)
$\langle S_1 \rangle$? [MASK] $\langle S_2 \rangle$	Yes/Maybe/No	74.9 (3.0)
$\langle S_1 \rangle$ $\langle S_2 \rangle$ [MASK]	Yes/Maybe/No	65.8 (2.4)
$\langle S_2 \rangle$? [MASK] , $\langle S_1 \rangle$	Yes/Maybe/No	62.9 (4.1)
$\langle S_1 \rangle$? [MASK] , $\langle S_2 \rangle$	Maybe/No/Yes	60.6 (4.8)
Fine-tuning	-	48.4 (4.8)

How to design good prompts?

- **BoolQ**: given a passage q and question p , design a prompt for question answering.

For **BoolQ**, given a passage p and question q :

p . Question: q ? Answer: <MASK>.

p . Based on the previous passage, q ?
<MASK>.

Based on the following passage, q ? <MASK>.
 p

with "yes" or "no" as verbalizers for True and False.

How to design good prompts? (cont.)

- **WiC**: given two sentences S_1 and S_2 , and a word W , design a prompt to determine whether W was used in the same sense in both sentences.

For **WiC**, given two sentences s_1 and s_2 and a word w , we classify whether w was used in the same sense.

" s_1 " / " s_2 ". Similar sense of " w "? <MASK>.

s_1 s_2 Does w have the same meaning in both sentences? <MASK>.

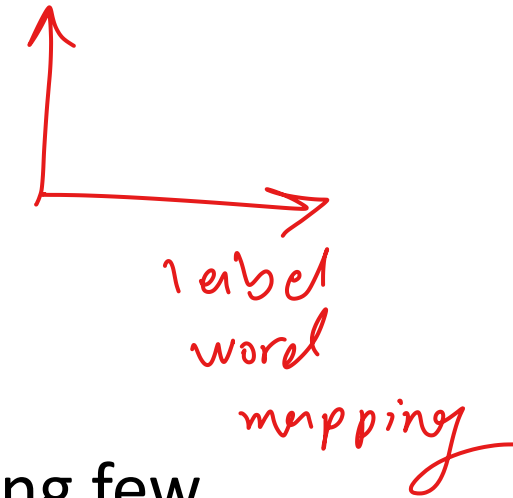
How to design good prompts? (cont.)

- Manual designing requires some **effort**.
- The template T and word-class mapping M are **not independent**.
- Model selection (T, M) is subject to **overfitting**.

CO-ordinated
- descent

~²
-

Template



Automatic Selection of Label Words

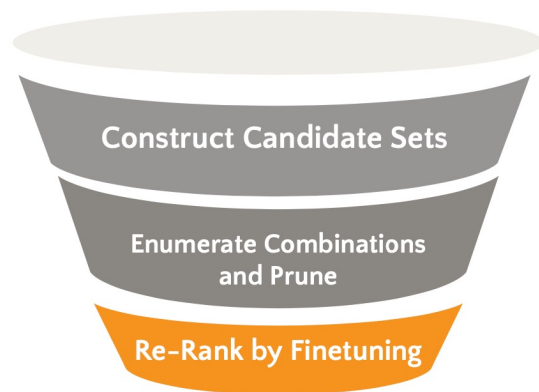
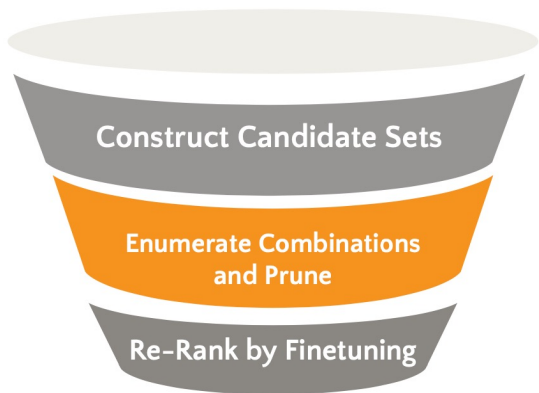
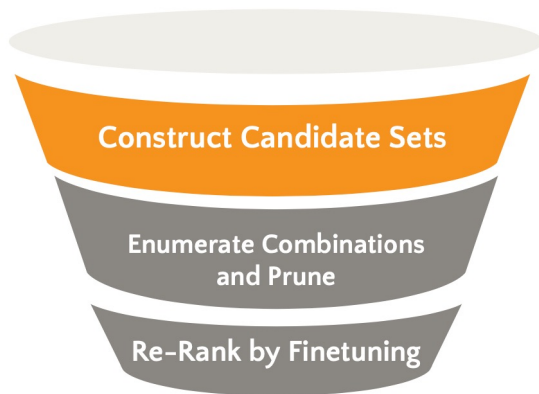
- Why naively searching all possibilities is not working?
- Generally **interactable**, exponentially large search space.
- Prone to overfitting. May uncover **spurious correlations** using few samples.
- For each **class c**, select **top k** words according to

$$\text{Top-}k_{v \in \mathcal{V}} \left\{ \sum_{x_{\text{in}} \in \mathcal{D}_{\text{train}}^c} \log P_{\mathcal{L}} \left([\text{MASK}] = v \mid \mathcal{T}(x_{\text{in}}) \right) \right\}$$

- $\mathcal{D}_{\text{train}}^c$ is training set for the class c.

Automatic Selection of Label Words (cont.)

- **Enumerate** all combinations of top-k words for different classes.
- **Prune** by zero-shot accuracy on the training set, select top-n tuples.
- **Fine-tune** based on top-n candidate and select the best one on the dev set.



Given the **manual** template: <S> It was [MASK] .

label:positive

good
great
perfect
...

label:negative

awful
bad
terrible
...

good/awful
good/bad
...
perfect/bad
perfect/terrible

Prune

good/bad
great/bad
great/terrible
perfect/terrible

Given the **manual** template: <S> It was [MASK] .

good/bad
great/bad
great/terrible
perfect/terrible

Fine-tune and
evaluate on $\mathcal{D}_{dev}$

good/bad (85%)
great/bad (82%)
great/terrible (91%)
perfect/terrible (86%)

Automatic Generation of Templates

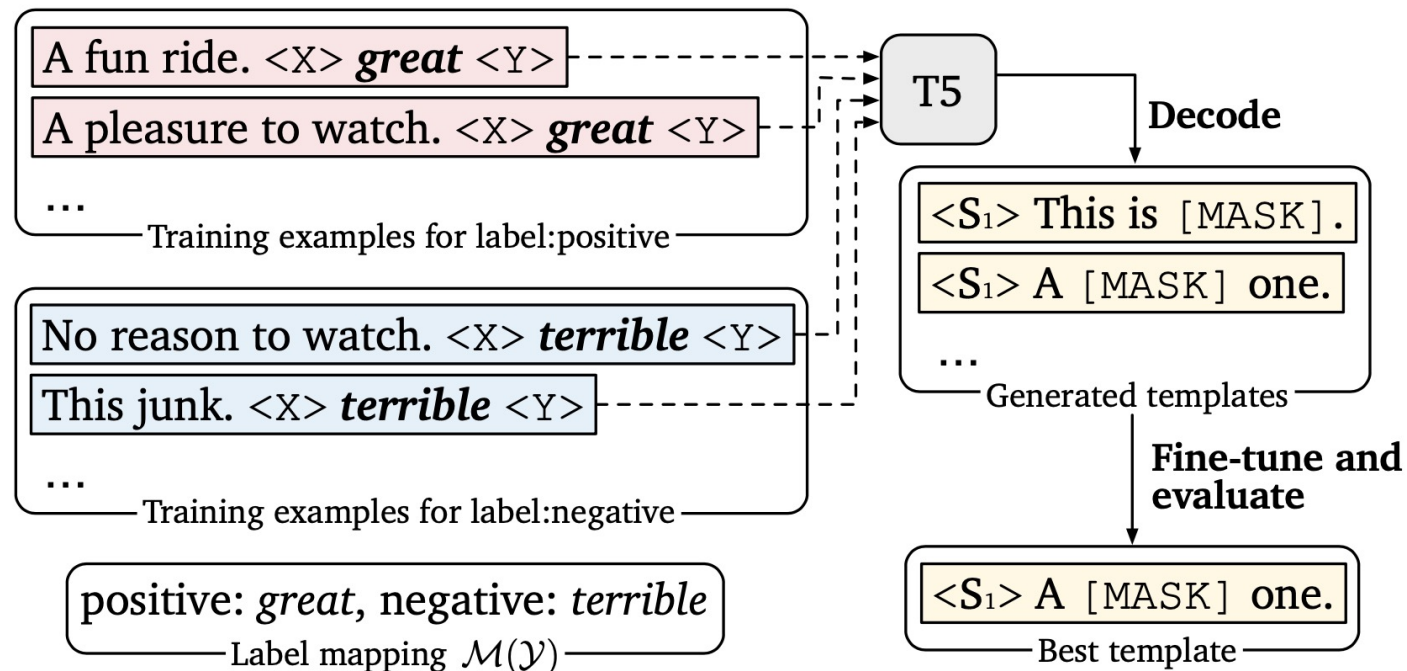
- Having **fixed $M(y)$** , use the T5 model.
 - Trained to fill in multiple tokens.
 - e.g. “Thank you $\langle X \rangle$ to your party $\langle Y \rangle$ week” with $X = \text{“inviting me”}$ and $Y = \text{“last”}$
- Let $T_g(x_{\text{in}}, y)$ be the formulation for making the **T5 input**:

$$\begin{aligned}\langle S_1 \rangle &\longrightarrow \langle X \rangle \mathcal{M}(y) \langle Y \rangle \langle S_1 \rangle, \\ \langle S_1 \rangle &\longrightarrow \langle S_1 \rangle \langle X \rangle \mathcal{M}(y) \langle Y \rangle, \\ \langle S_1 \rangle, \langle S_2 \rangle &\longrightarrow \langle S_1 \rangle \langle X \rangle \mathcal{M}(y) \langle Y \rangle \langle S_2 \rangle.\end{aligned}$$

$$\begin{aligned}&\sum_{(x_{\text{in}}, y) \in \mathcal{D}_{\text{train}}} \log P_{\text{T5}}(\mathcal{T} \mid \mathcal{T}_g(x_{\text{in}}, y)) \\ &\sum_{j=1}^{|\mathcal{T}|} \sum_{(x_{\text{in}}, y) \in \mathcal{D}_{\text{train}}} \log P_{\text{T5}}(t_j \mid t_1, \dots, t_{j-1}, \mathcal{T}_g(x_{\text{in}}, y))\end{aligned}$$

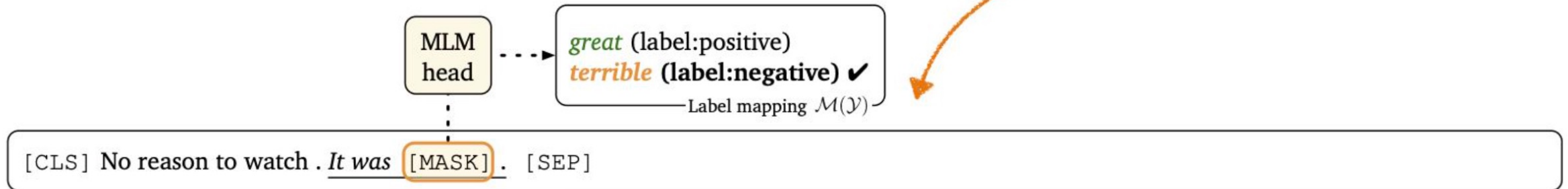
Automatic Generation of Templates (cont.)

- Use a wide ($b = 100$) **beam search** to decode $\langle X \rangle$ and $\langle Y \rangle$.
- Finally, fine-tune the model on top-p templates and pick the one with best dev accuracy. c



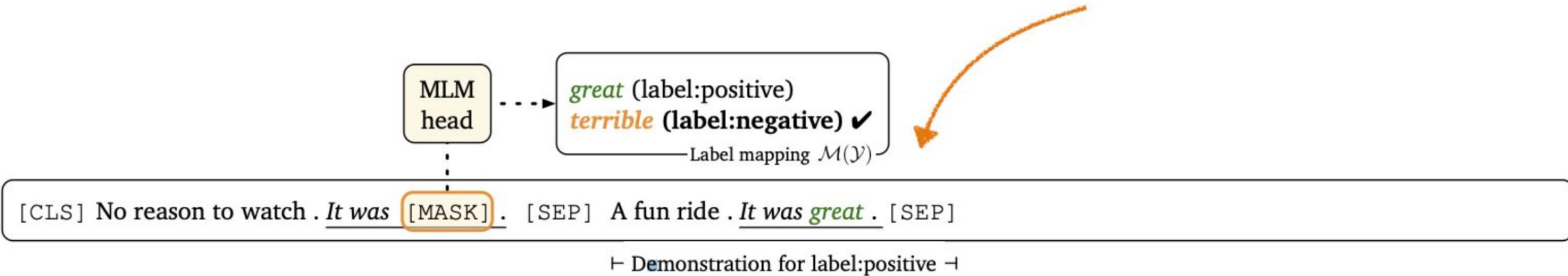
Demonstrations

GPT3 In-context Learning:
Randomly Samples Examples and fills
them in context 🖐️



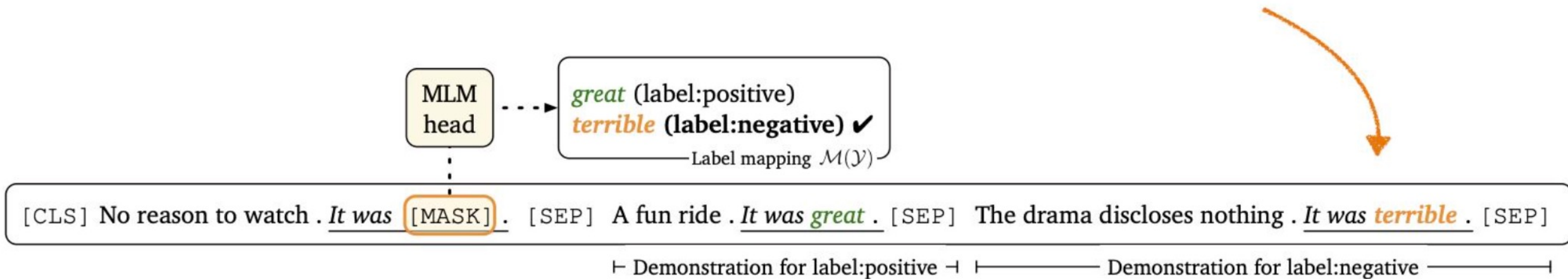
Demonstrations (cont.)

Improved: Selective Sampling, ie. for this example sample from then positive class 😎



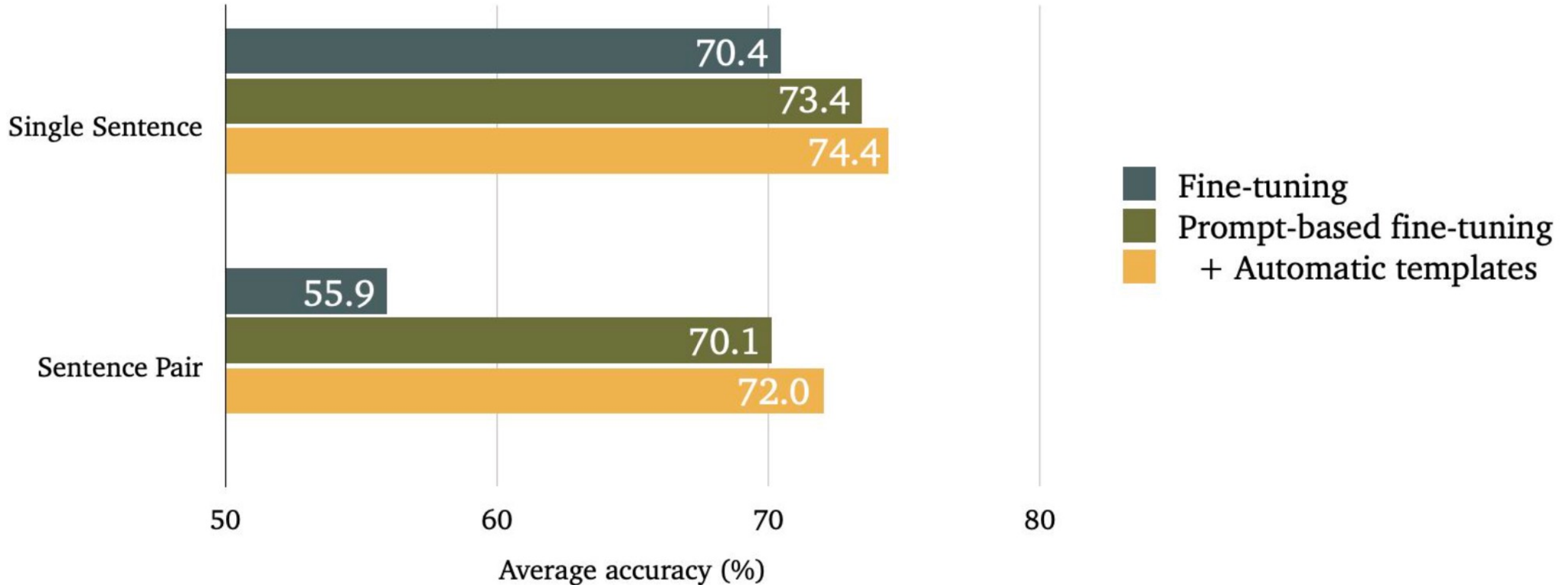
Demonstrations (cont.)

And we can also sample one from a negative training instance



- How to select demo samples?

Ablation Studies



Ablation Studies (cont.)

